INTAKE-ARBITER -- FREE-COOLING DECISION REPORT

Data centres over-cool, continuously, because nobody can promise them the hours ahead. A chiller plant needs hours of notice to change mode, and a thermometer only ever reports NOW -- so the mechanical chillers keep running through hours that outside air could have cooled for free. FortyGuard closes that gap with heat intelligence 2 m above the ground, the height a ground-mounted condenser actually breathes. This agent turns that forecast into an hour-by-hour schedule carrying a calibrated safety bound.

THE SITE

Location	Amazon Web Services, VA -- station KDAA, America/New_York
Building	OSM 1536981894 Amazon Web Services
Facade gap	not applicable -- one building, no second facade
Weather record	42,250 real hourly records from KDAA
Plume physics	NOT MODELLED. No other tagged data centre lies inside the solver's validated range, so there is no neighbour intake for a plume to arrive at: the quantity does not exist here rather than being unmeasured. This is a statement about the model's domain, not a claim that recirculation here is zero. A building's own exhaust re-entering its own intake is not modelled at any site in this project.

FortyGuard data: weather and geometry are this site's own; the four measured FortyGuard level offsets and the N-26 coverage record are Ashburn's, because only Ashburn has forecast/outcome day pairs. Quote the hours for this site; quote the coverage for Ashburn.

THIS REPORT IS A SNAPSHOT, NOT THE LIVE PAGE

The interface recomputes the decision for whatever configuration a reader selects. This file was generated at build time for the ONE configuration named below, chosen because it is the configuration in which this site's agent does the most while still changing mode at least once. If the numbers on screen differ from the numbers here, the configuration differs -- compare the two lists before concluding anything else.

Day	2023-06-12 (chatter)
Plant limit	21.0 C
Notice required	3 h
Forecast skill	0.50 relative to persistence
Level anchor	sensor
Condenser bank	longest facade
Switch budget	2 mode changes per day
Minimum dwell	3 h
Max dew point	15.0 C (Green Grid WP#46 p.6)

WHAT THE AGENT DID ON THIS DAY

The agent ran free cooling for 1 of 23 hours with 1 mode change. That is 1 more than the reactive incumbent, at 0 unsafe hours against its 0.

Agent	1 of 23 hours free cooling, 1 mode change(s), 0 unsafe hour(s)
Incumbent	0 of 23 hours free cooling, 0 mode change(s), 0 unsafe hour(s)

EVERY HOUR, AND THE REASON FOR IT

hh	mode	safe	binding	bound	limit	actual	margin
00	mechanical	no	dry-bulb	24.360	21.0	24.220	0.949
01	mechanical	no	dry-bulb	24.250	21.0	23.720	0.911
02	mechanical	no	dry-bulb	24.110	21.0	23.330	0.872
04	mechanical	no	dew point	23.195	21.0	21.390	0.840
05	mechanical	no	dew point	22.695	21.0	20.780	0.848
06	mechanical	no	dew point	23.275	21.0	22.000	0.851

07	mechanical	no	dew point	23.835	21.0	22.390	1.447
08	mechanical	no	dry-bulb	26.415	21.0	24.440	2.067
09	mechanical	no	dry-bulb	28.055	21.0	24.000	2.437
10	mechanical	no	dry-bulb	28.610	21.0	24.000	2.686
11	mechanical	no	dry-bulb	28.195	21.0	23.780	1.957
12	mechanical	no	dry-bulb	27.165	21.0	24.440	1.316
13	mechanical	no	dry-bulb	26.110	21.0	23.610	1.098
14	mechanical	no	dew point	25.115	21.0	22.780	0.975
15	mechanical	no	dew point	25.165	21.0	23.110	0.926
16	mechanical	no	dew point	25.140	21.0	24.780	1.006
17	mechanical	no	dew point	24.390	21.0	25.000	1.200
18	mechanical	no	dew point	22.390	21.0	21.670	1.369
19	mechanical	no	dew point	23.250	21.0	22.390	1.514
20	mechanical	no	dew point	22.970	21.0	21.500	1.764
21	mechanical	no	dew point	20.975	21.0	20.560	1.668
22	mechanical	no	dew point	21.250	21.0	20.220	1.360
23	FREE	yes	-	20.280	21.0	18.890	1.112

bound = the agent's 90 %-nominal upper bound on intake air, forecast plus margin
actual = what the intake temperature turned out to be, from the station record
margin = group-conditional forecast error for that hour of day, plus plume spread

THE REASONING, HOUR BY HOUR

00:00 MECHANICAL [dry-bulb]
Mechanical. The upper bound on intake air is 24.360 C against a 21.0 C limit, so it fails by 3.360 C. It would take a limit 3.360 C higher, or a bound 3.360 C tighter, to change this hour.

01:00 MECHANICAL [dry-bulb]
Mechanical. The upper bound on intake air is 24.250 C against a 21.0 C limit, so it fails by 3.250 C. It would take a limit 3.250 C higher, or a bound 3.250 C tighter, to change this hour.

02:00 MECHANICAL [dry-bulb]
Mechanical. The upper bound on intake air is 24.110 C against a 21.0 C limit, so it fails by 3.110 C. It would take a limit 3.110 C higher, or a bound 3.110 C tighter, to change this hour.

04:00 MECHANICAL [dew point]
Mechanical, and TEMPERATURE IS NOT THE REASON -- the dry-bulb bound of 23.195 C would have passed. The outside air is too HUMID: the dew-point bound is 17.25 C against a 15.0 C maximum, failing by 2.25 C. Cool but damp air condenses on cold surfaces inside the hall, which is why real economizers gate on humidity and not on temperature alone.

05:00 MECHANICAL [dew point]
Mechanical, and TEMPERATURE IS NOT THE REASON -- the dry-bulb bound of 22.695 C would have passed. The outside air is too HUMID: the dew-point bound is 17.55 C against a 15.0 C maximum, failing by 2.55 C. Cool but damp air condenses on cold surfaces inside the hall, which is why real economizers gate on humidity and not on temperature alone.

06:00 MECHANICAL [dew point]
Mechanical, and TEMPERATURE IS NOT THE REASON -- the dry-bulb bound of 23.275 C would have passed. The outside air is too HUMID: the dew-point bound is 18.26 C against a 15.0 C maximum, failing by 3.26 C. Cool but damp air condenses on cold surfaces inside the hall, which is why real economizers gate on humidity and not on temperature alone.

07:00 MECHANICAL [dew point]
Mechanical, and TEMPERATURE IS NOT THE REASON -- the dry-bulb bound of 23.835 C would have passed. The outside air is too HUMID: the dew-point bound is 19.11 C against a 15.0 C maximum, failing by 4.11 C. Cool but damp air condenses on cold surfaces inside the hall, which is why real economizers gate on humidity and not on temperature alone.

08:00 MECHANICAL [dry-bulb]
Mechanical. The upper bound on intake air is 26.415 C against a 21.0 C limit, so it

fails by 5.415 C. It would take a limit 5.415 C higher, or a bound 5.415 C tighter, to change this hour.

09:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 28.055 C against a 21.0 C limit, so it fails by 7.055 C. It would take a limit 7.055 C higher, or a bound 7.055 C tighter, to change this hour.

10:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 28.610 C against a 21.0 C limit, so it fails by 7.610 C. It would take a limit 7.610 C higher, or a bound 7.610 C tighter, to change this hour.

11:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 28.195 C against a 21.0 C limit, so it fails by 7.195 C. It would take a limit 7.195 C higher, or a bound 7.195 C tighter, to change this hour.

12:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 27.165 C against a 21.0 C limit, so it fails by 6.165 C. It would take a limit 6.165 C higher, or a bound 6.165 C tighter, to change this hour.

13:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 26.110 C against a 21.0 C limit, so it fails by 5.110 C. It would take a limit 5.110 C higher, or a bound 5.110 C tighter, to change this hour.

14:00 MECHANICAL [dew point]

Mechanical, and TEMPERATURE IS NOT THE REASON -- the dry-bulb bound of 25.115 C would have passed. The outside air is too HUMID: the dew-point bound is 20.89 C against a 15.0 C maximum, failing by 5.89 C. Cool but damp air condenses on cold surfaces inside the hall, which is why real economizers gate on humidity and not on temperature alone.

15:00 MECHANICAL [dew point]

Mechanical, and TEMPERATURE IS NOT THE REASON -- the dry-bulb bound of 25.165 C would have passed. The outside air is too HUMID: the dew-point bound is 21.30 C against a 15.0 C maximum, failing by 6.30 C. Cool but damp air condenses on cold surfaces inside the hall, which is why real economizers gate on humidity and not on temperature alone.

16:00 MECHANICAL [dew point]

Mechanical, and TEMPERATURE IS NOT THE REASON -- the dry-bulb bound of 25.140 C would have passed. The outside air is too HUMID: the dew-point bound is 21.73 C against a 15.0 C maximum, failing by 6.73 C. Cool but damp air condenses on cold surfaces inside the hall, which is why real economizers gate on humidity and not on temperature alone.

17:00 MECHANICAL [dew point]

Mechanical, and TEMPERATURE IS NOT THE REASON -- the dry-bulb bound of 24.390 C would have passed. The outside air is too HUMID: the dew-point bound is 20.86 C against a 15.0 C maximum, failing by 5.86 C. Cool but damp air condenses on cold surfaces inside the hall, which is why real economizers gate on humidity and not on temperature alone.

18:00 MECHANICAL [dew point]

Mechanical, and TEMPERATURE IS NOT THE REASON -- the dry-bulb bound of 22.390 C would have passed. The outside air is too HUMID: the dew-point bound is 21.55 C against a 15.0 C maximum, failing by 6.55 C. Cool but damp air condenses on cold surfaces inside the hall, which is why real economizers gate on humidity and not on temperature alone.

19:00 MECHANICAL [dew point]

Mechanical, and TEMPERATURE IS NOT THE REASON -- the dry-bulb bound of 23.250 C would have passed. The outside air is too HUMID: the dew-point bound is 20.53 C against a 15.0 C maximum, failing by 5.53 C. Cool but damp air condenses on cold surfaces inside the hall, which is why real economizers gate on humidity and not on temperature alone.

20:00 MECHANICAL [dew point]

Mechanical, and TEMPERATURE IS NOT THE REASON -- the dry-bulb bound of 22.970 C would have passed. The outside air is too HUMID: the dew-point bound is 17.39 C against a 15.0 C maximum, failing by 2.39 C. Cool but damp air condenses on cold surfaces inside the hall, which is why real economizers gate on humidity and not on temperature alone.

21:00 MECHANICAL [dew point]

Mechanical, and TEMPERATURE IS NOT THE REASON -- the dry-bulb bound of 20.975 C would have passed. The outside air is too HUMID: the dew-point bound is 16.70 C against a 15.0 C maximum, failing by 1.70 C. Cool but damp air condenses on cold surfaces inside the hall, which is why real economizers gate on humidity and not on temperature alone.

22:00 MECHANICAL [dew point]

Mechanical, and TEMPERATURE IS NOT THE REASON -- the dry-bulb bound of 21.250 C would have passed. The outside air is too HUMID: the dew-point bound is 15.50 C against a 15.0 C maximum, failing by 0.50 C. Cool but damp air condenses on cold surfaces inside the hall, which is why real economizers gate on humidity and not on temperature alone.

23:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 20.280 C, which is 0.720 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 1.112 C, and the margin is measured, not chosen: 1.112 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

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